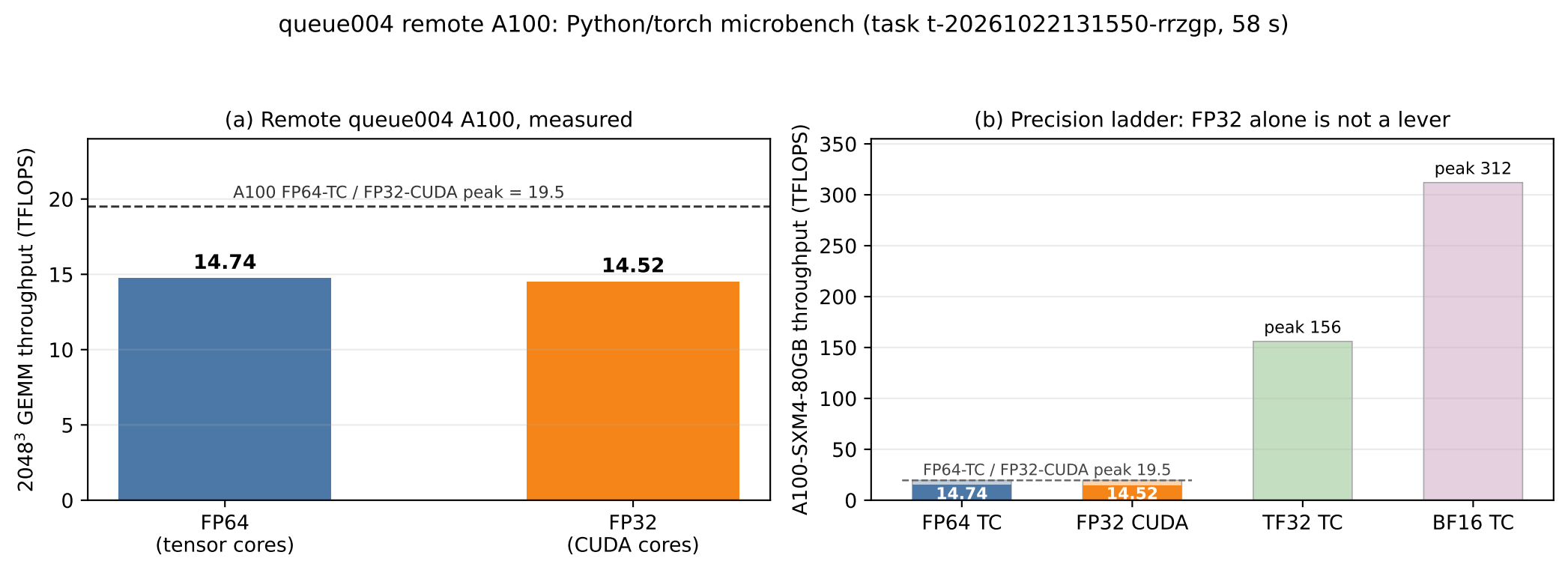
queue004 remote A100: Python/torch microbench (task t-20261022131550-rrzgp, 58 s)
(a) Remote queue004 A100, measured
2048³ GEMM throughput (TFLOPS)
A100 FP64-TC / FP32-CUDA peak = 19.5
14.74
14.52
FP64
(tensor cores)
FP32
(CUDA cores)
(b) Precision ladder: FP32 alone is not a lever
A100-SXM4-80GB throughput (TFLOPS)
FP64-TC / FP32-CUDA peak 19.5
14.74
14.52
peak 156
peak 312
FP64 TC
FP32 CUDA
TF32 TC
BF16 TC
torch 2.5.1+cu121, TF32 off (default), 2048³ GEMM, 20 iters; FP32 is 1.02x the FP64 wall time (no speedup)
Filled bars = measured FP64/FP32; faint bars = theoretical A100-SXM4-80GB peaks. TF32/BF16 require far more precision relaxation than FP32.